Supplementary Information (Anjur-Dietrich, *et al.*).

Supplementary Table 1. Time in generations to biofilm enrichments in *Prochlorococcus* strains. Ecotype abbreviations: HL = High-light adapted, LL = low-light adapted.

Strain	Ecotype	Generations to biofilm
MED4	HL	17.8
9301	HL	16.1
9312	HL	9.7
1205	LL	51.6
9313	LL	35.0
9211	LL	12.1
NATL2A	LL	23.0

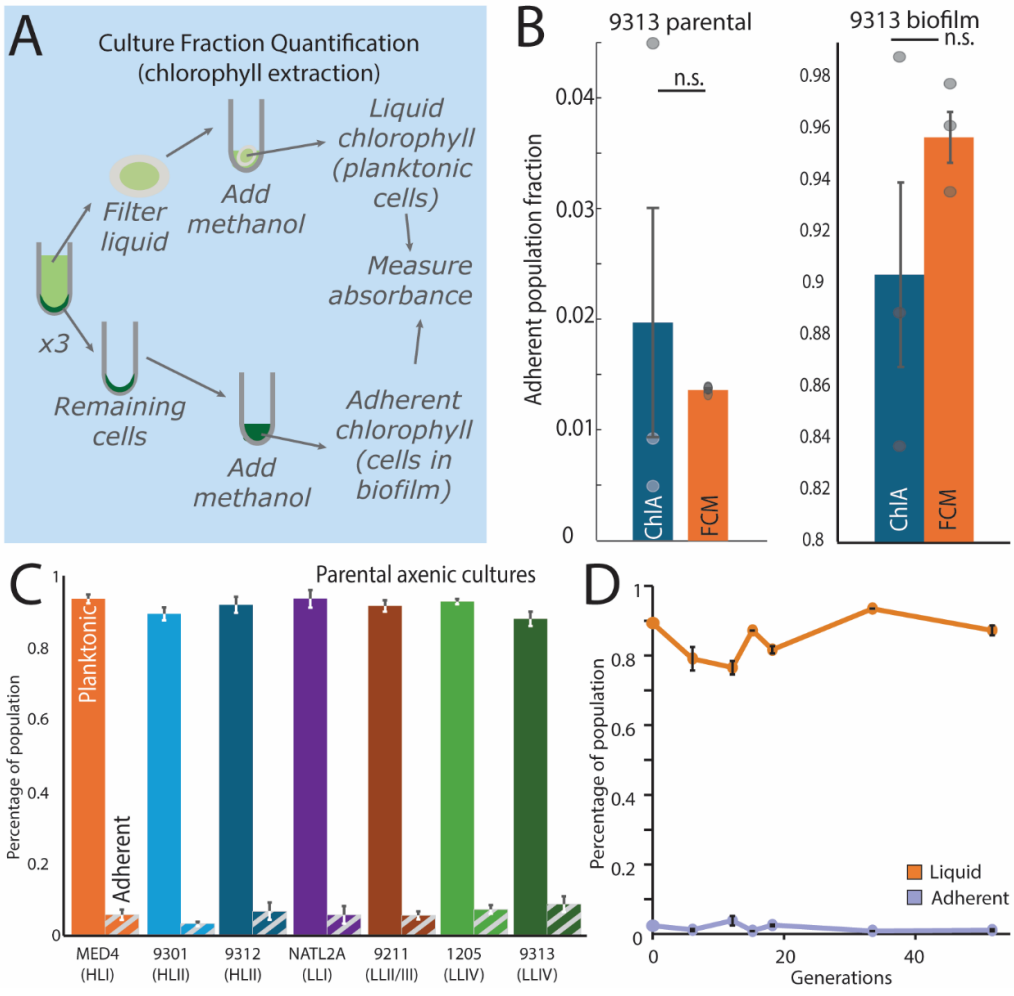
Supplementary Table 2. *Prochlorococcus* fractions of the total microbial community, measured by normalized metagenomic reads. The aggregated metagenomic collection includes samples from HOT346 (collected for this study), TARA Oceans, and samples described by Leu, *et al.* 2022 from the Hawai'i Ocean Time series (see Supp. Data 1 for identifiers). We define "free-living" as cells collected in the 0.2-1.2 μm size fraction and "particle-bound" as the >=1.2 μm size fraction. The combined euphotic zone is defined as 5-150 m, while the combined mesopelagic zone is defined as 300-1000 m. P-values were calculated using a two-sided t-test to account for unequal sample sizes.

Depth (m)	Free-living (%)	Particle-bound (%)	p-value	N, Free-living	N, Particle-bound
5	63±20	12±14	<<0.001	67	28
45	74±12	3±0	n/a	71	1
75	70±13	15±7	<<0.001	42	16
150	46±24	5±3	<<0.001	108	21
300	5±6	1±0.9	0.04	102	11
1000	1±0.3	3±5	0.0008	93	10
Euphotic	60±22	11±11	<<0.001	288	66
Mesopelagic	3±5	2±3	0.25 (n.s.)	195	21

Supplementary Table 3. Clade breakdown of *Prochlorococcus* populations reported in Supp. Table 2. HL = High-light adapted, LL = low-light adapted. The combined euphotic zone is defined as 5-150 m, while the combined mesopelagic zone is defined as 300-1000 m. P-values were calculated using a two-sided t-test to account for unequal sample sizes.

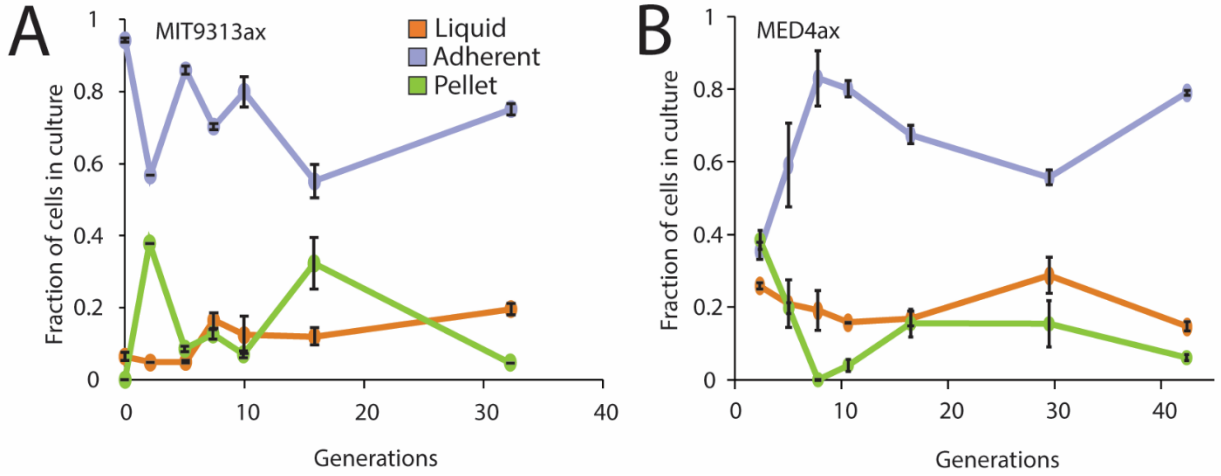
Free-living			
Depth (m)	**HL %**	**LL %**	**N**
5	84±11	3±11	67
45	87±9	2±9	71
75	83±11	5±11	42
150	33±31	59±33	108
300	17±13	74±14	96
1000	68±9	21±9	19
Euphotic	65±33	24±34	288
Mesopelagic	25±23	65±24	115
Particle-bound			
Depth (m)	**HL %**	**LL %**	**N**
5	70±15	10±12	28
45	54±0	23±0	1
75	75±9	8±6	16
150	42±36	46±37	21
0.92 (300	43±21	44±22	10
1000	62±22	26±26	9
Euphotic	62±27	21±28	66
Mesopelagic	52±23	36±25	19

Supplementary Figure 1. Parental *Prochlorococcus* populations consistently have a small percentage of adherent cells, whether measured by cell number or chlorophyll. (A) At each timepoint, the liquid fraction (i.e. media and planktonic cells) was removed and filtered, and then the filter was incubated in methanol. Fresh methanol was added to the original culture tube and vortexed to dissolve any adherent cells. Both samples were incubated at 4 C for 18 hours, and then the percentage of planktonic and adherent cells (in the liquid and adherent fractions, respectively) was measured as total extracted chlorophyll using standard techniques. **(B)** Population fractions of liquid and adherent *Prochlorococcus* measured by chlorophyll absorption or cell counting. The percentage of adherent cells in parental cultures of MIT9313 was 2.0 ± 1.0% measured by chlorophyll absorption and 1.4 ± 0.00015% measured by flow cytometry (n=3, p=0.66). In biofilm populations, 90.4 ± 3.6% of cells were adherent measured by chlorophyll absorption and 95.7 ± 1.0% by flow cytometry (n=3, p=0.31). (For B, C, D: All n reported in this figure are biological replicates. All data are reported as mean ± SEM, and p-values are calculated using a 2-sided t-test.) **(C)** Liquid and adherent population fractions in parental axenic cultures. Across clades, parental strains were composed of 92 ± 0.7% planktonic cells and 6 ± 0.6% biofilm cells (n>= 10 for all strains). There was no significant difference between the fraction of adherent cells in high-light and low-light adapted clades (5 ± 0.9% and 7 ± 0.6%, respectively, p=0.13). **(D)** Population fractions of liquid and adherent *Prochlorococcus* in a planktonic enrichment of a low-light strain (MIT9313) (n=3).

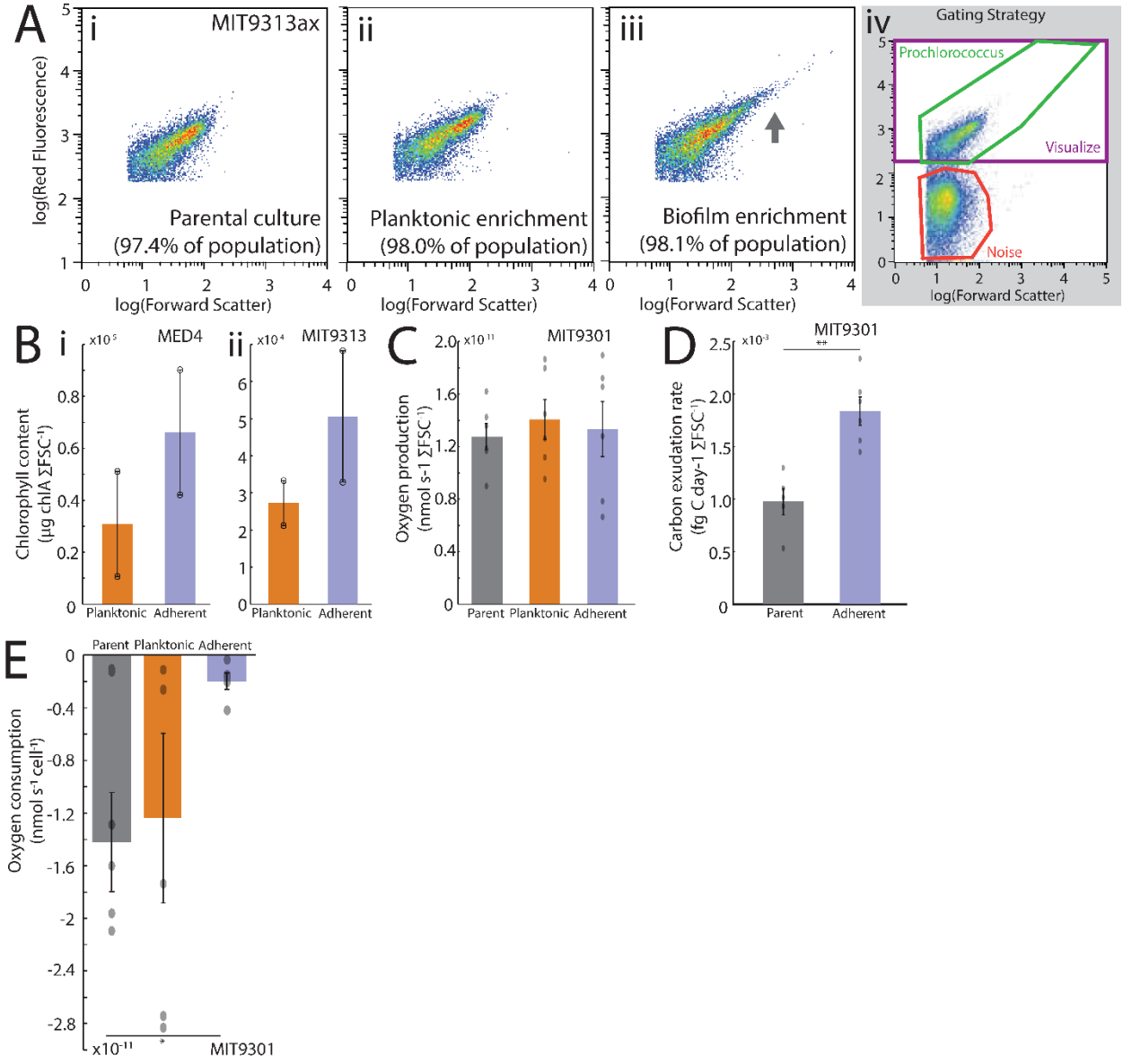


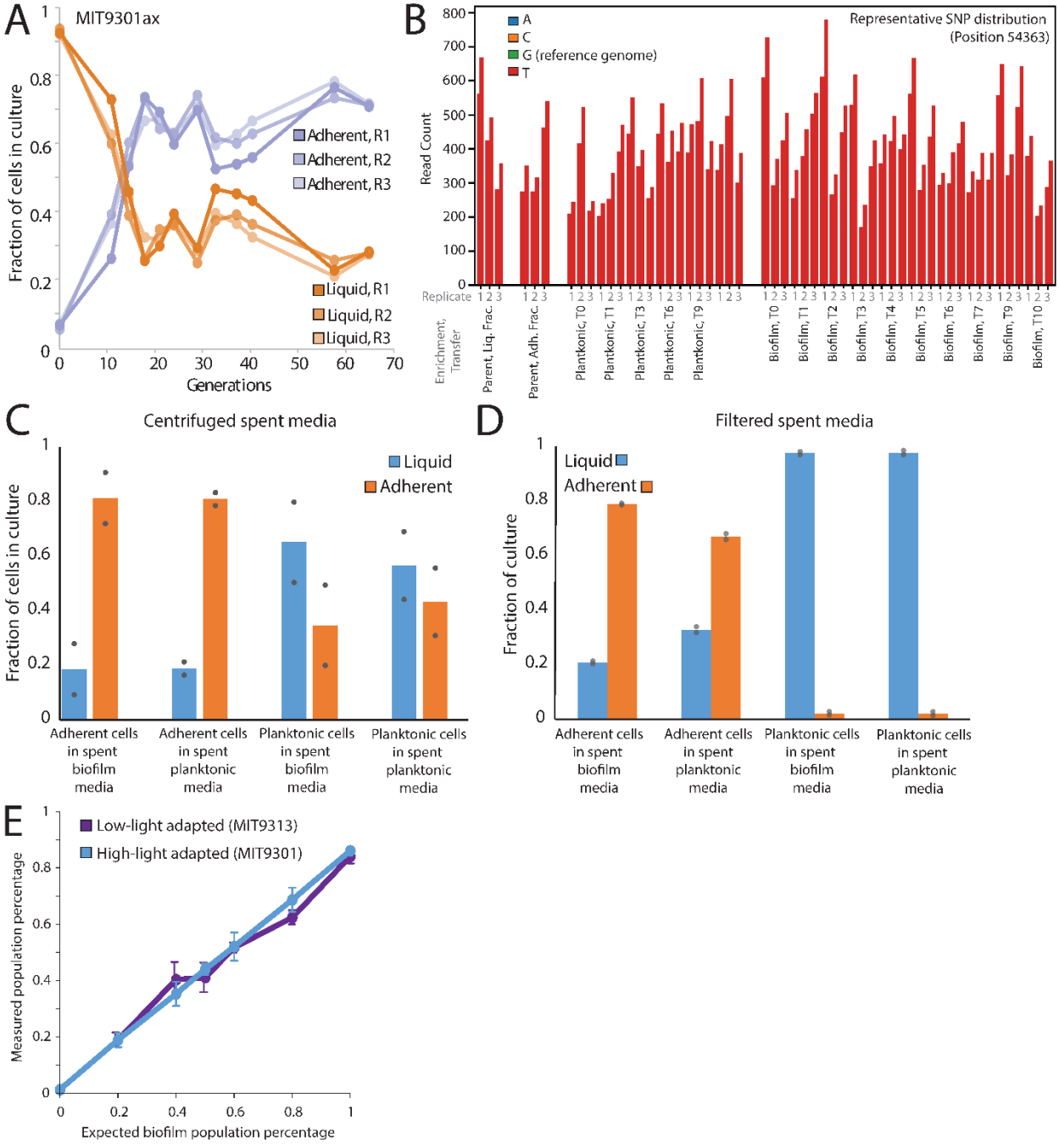
Supplementary Figure 2. Daily physical perturbation of biofilms is insufficient to return cultures to a planktonic state. (A) Fractions of a biofilm population from a low-light adapted strain (MIT9313ax) that was vortexed daily. At each transfer, the entire culture was resuspended (rather than isolating only adherent cells as during biofilm enrichment) and used to inoculate a new culture. (For A and B: n=3 biological replicates at each timepoint, data are presented as mean ± SEM). **(B)** Fractions of a biofilm population from a high-light adapted strain (MED4ax) that was vortexed daily (same experimental setup as described in A).

We note that, unlike in other experiments, vortexing cultures daily resulted in large loose pellets, which formed while the population was growing exponentially (green line). When enriched, pellet populations eventually formed biofilm populations with no evidence of pellets (data not shown).



Supplementary Figure 3. Biofilm formation is repeatable, non-genetic, and not due to a factor found in spent media. (A) Population fractions measured by chlorophyll *a* absorption from replicate enrichments for biofilms. **(B)** Base calls from a representative position of parental, planktonic enrichment, and biofilm enrichment populations of a high-light strain (MIT9301) mapped onto a closed reference genome (enrichments each selected for over 100 generations). **(C)** Fraction of planktonic and adherent cells in cultures with cells either from planktonic or biofilm cultures grown in centrifuged, cell-free spent media from either planktonic or biofilm cultures (MIT9301). Cultures were grown for 5 days, and cells were collected in exponential phase. (For C and D: n=2 biological replicates, data are presented as mean only). **(D)** Fraction of planktonic and adherent cells in cultures using the same experimental setup as in (C) with the exception that the spent media was 0.2 μm-filtered (MIT9301). **(E)** Initial inoculation ratio of adherent to planktonic cells compared to measured adherent fraction after 3 days of growth (n=3 biological replicates, data are presented as mean ± SEM).


A
i
MIT9313ax
log(Red Fluorescence)
log(Forward Scatter)
Parental culture
(97.4% of population)
ii
Planktonic enrichment
(98.0% of population)
iii
Biofilm enrichment
(98.1% of population)
iv
Gating Strategy
Prochlorococcus
Visualize
Noise
log(Forward Scatter)
B
i
MED4
Chlorophyll content
(µg chlA ΣFSC-1)
Planktonic
Adherent
ii
MIT9313
C
MIT9301
Oxygen production
(nmol s-1 ΣFSC-1)
Parent
Planktonic
Adherent
D
MIT9301
Carbon exudation rate
(fg C day-1 ΣFSC-1)
Parent
Adherent
E
Oxygen consumption
(nmol s-1 cell-1)
MIT9301

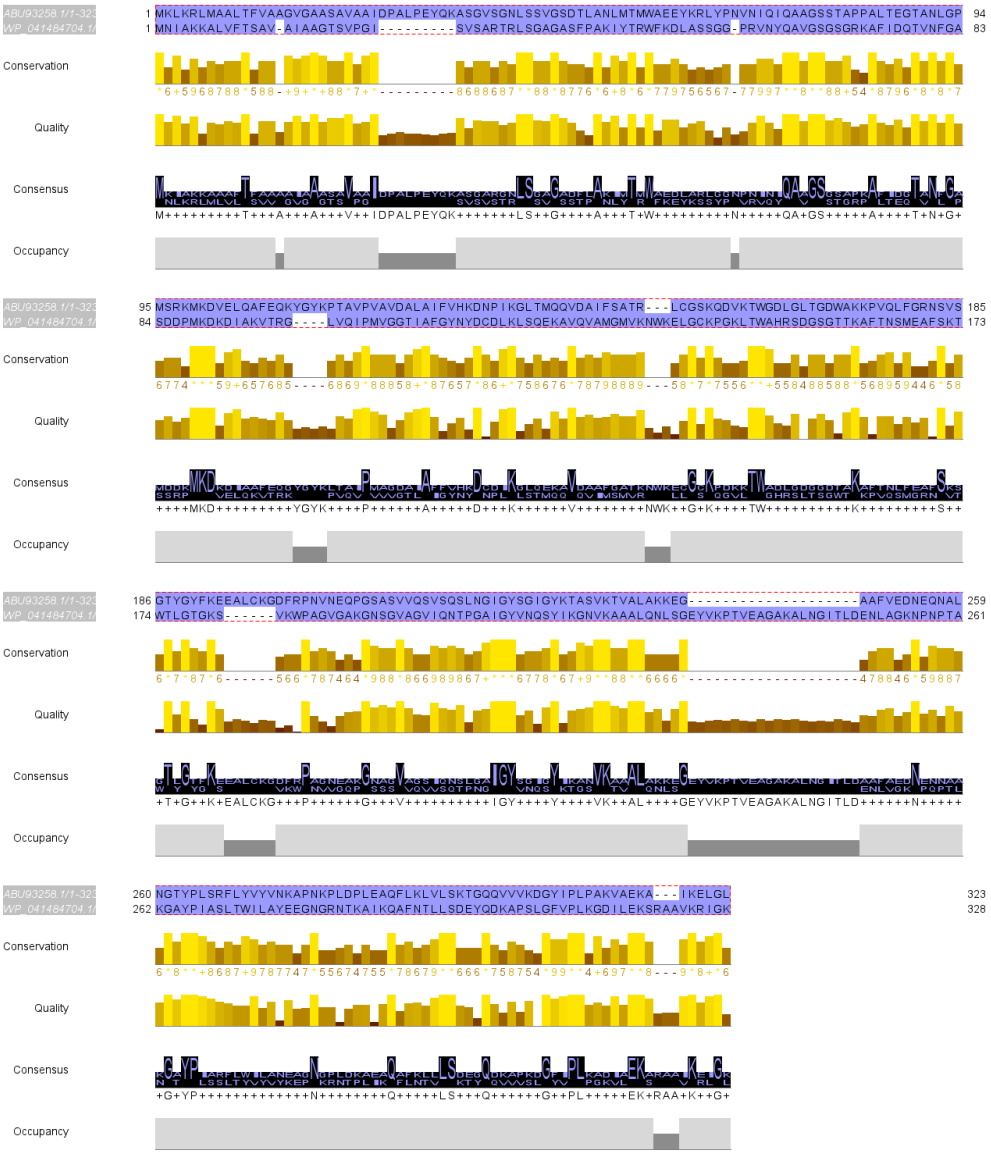

A
MIT9301ax
Fraction of cells in culture
Generations
Adherent, R1
Adherent, R2
Adherent, R3
Liquid, R1
Liquid, R2
Liquid, R3
B
Representative SNP distribution
(Position 54363)
A
C
G (reference genome)
T
Read Count
Replicate
Enrichment, Transfer
Parent, Liq. Frac.
Parent, Adh. Frac.
Planktonic, T0
Planktonic, T1
Planktonic, T3
Planktonic, T6
Planktonic, T9
Biofilm, T0
Biofilm, T1
Biofilm, T2
Biofilm, T3
Biofilm, T4
Biofilm, T5
Biofilm, T6
Biofilm, T7
Biofilm, T9
Biofilm, T10
C
Centrifuged spent media
Liquid
Adherent
Fraction of cells in culture
Adherent cells in spent biofilm media
Adherent cells in spent planktonic media
Planktonic cells in spent biofilm media
Planktonic cells in spent planktonic media
D
Filtered spent media
Liquid
Adherent
Fraction of culture
Adherent cells in spent biofilm media
Adherent cells in spent planktonic media
Planktonic cells in spent biofilm media
Planktonic cells in spent planktonic media
E
Low-light adapted (MIT9313)
High-light adapted (MIT9301)
Measured population percentage
Expected biofilm population percentage

66

67

Supplementary Figure 4. Additional measurements supporting carbon flux tracking in parental and biofilm populations. (A) Representative flow cytograms of parental (i), planktonic enrichment (ii), and biofilm enrichment (iii) populations. The grey arrow in panel (iii) highlights the elongated tail in forward scatter seen in adherent cells. The gating strategy for all flow cytograms is presented in (iv). (B) Chlorophyll content normalized to integrated forward scatter (ΣFSC) as a proxy for total MED4 population biovolume (i) and total MIT9313 population biovolume (ii). Note that there are only 2 replicates for MIT9313, since 1 replicate presented in the main text did not have a bead standard in the flow cytometry run. (C) Oxygen production rate normalized to integrated forward scatter, measured in MIT9301. (D) Carbon exudation rate normalized to integrated forward scatter, measured in MIT9301. (E) Respiration rates for parental, planktonic, and biofilm populations of axenic MIT9301 measured in darkness: parent: 1.42×10^{-11} nmol O_2 s^{-1} $cell^{-1}$, n=7; planktonic: 1.24×10^{-11} nmol O_2 s^{-1} $cell^{-1}$, n=4; biofilm: 2.0×10^{-12} nmol O_2 s^{-1} $cell^{-1}$, n=5. Note that due to the variability in the planktonic measurements, there is no significant difference between the planktonic oxygen evolution rates and either the parental or biofilm rates.

Supplementary Figure 5. The PstS amino acid sequences from *Pseudomonas aeruginosa* and *Prochlorococcus* are in strong agreement. Alignment of *Pseudomonas aeruginosa* (NP_254056.1) and *Prochlorococcus marinus* (WP_041484704.1) PstS amino acid sequence with an 87% query cover, $8x10^{-7}$ e-value, and 24.45% identity. Conservation indicates how closely the properties of the aligned amino acids match (top score 11 marked as *, score 10 marked as +). Quality is normalized from 0 to 1 (best) across the entire alignment, and then the consensus sequence and occupancy (identifying any alignment gaps) are calculated. (Visualized using Jalview)



Supplementary Figure 6. Additional measurements supporting experiments measuring cell emergence from biofilms. (A) Schematic describing experimental setup to track planktonic cells removed from biofilm cultures during the media swap (see also Fig. 4A). On day 3, when we replaced the media from the biofilm cultures, we transferred the planktonic cells to a new culture tube. Over the next 4 timepoints, we continued to remove the planktonic fraction, move it to a new culture tube, and thus discard any adhered cells. **(B)** Population counts for liquid and adherent fractions in the planktonic population (n=3, data are presented as mean ± SEM). (All n presented in this figure are biological replicates) **(C)** Chitin beads were incubated with MIT9301ax cells from either planktonic-enriched or biofilm-enriched cultures, and the cell concentrations in different culture fractions were quantified during mid-exponential growth. First, the cell concentration in the media in each culture was measured ("initial liquid fraction"). Planktonic cultures with chitin beads had $4.0x10^{8}\pm7.0x10^{6}$ cells mL^{-1} in the liquid phase, while biofilm cultures were about 100-time less concentrated with $5.5x10^{6}\pm3.1x10^{5}$ cells mL^{-1} (n=6). Next, the chitin beads were repeatedly washed with fresh media, and we measured the concentration of cells that washed off the beads in either culture ("washed off"). Slightly more cells were washed off chitin beads incubated with planktonic cultures ($3.3x10^{5}\pm1.5x10^{4}$ cells mL^{-1} compared to $2.7x10^{5}\pm3.6x10^{3}$ cells mL^{-1}, n=6). Finally, a subset of beads was vortexed for 3 minutes to dislodge all cells that remained attached to the chitin in either culture ("vortexed off"), and beads incubated with biofilm populations had a significantly higher number of attached cells compared to beads incubated with planktonic populations ($9.9x10^{4}\pm2.4x10^{3}$ cells mL^{-1} and $2.5x10^{4}\pm1.6x10^{4}$ cells mL^{-1}, respectively, n=3, p=0.05, 2-sided t-test). We first lightly washed the chitin beads to ensure that we would only count truly attached cells as "on" chitin beads, due to the porous texture of chitin beads which could trap non-adherent cells.

